

Tyler LaBonte

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Education

GEORGIA INSTITUTE OF TECHNOLOGY 2021–2026
Ph.D., Machine Learning

UNIVERSITY OF SOUTHERN CALIFORNIA 2017–2021
B.S., Applied and Computational Mathematics, *magna cum laude*

Skills: Python, PyTorch, TensorFlow, HuggingFace, DeepSpeed, W&B, Linux CLI, Kubernetes, Docker, Git, Vim, $\text{\LaTeX}$

Selected Publications

1. [Task Shift: From Classification to Regression in Overparameterized Linear Models](#)
Tyler LaBonte*, Kuo-Wei Lai*, and Vidya Muthukumar (* denotes equal contribution)
AISTATS 2025
2. [The Group Robustness is in the Details: Revisiting Finetuning under Spurious Correlations](#)
Tyler LaBonte, John C. Hill, Xincheng Zhang, Vidya Muthukumar, and Abhishek Kumar
NeurIPS 2024
3. [Towards Last-layer Retraining for Group Robustness with Fewer Annotations](#)
Tyler LaBonte, Vidya Muthukumar, and Abhishek Kumar
NeurIPS 2023
4. [Scaling Novel Object Detection with Weakly Supervised Detection Transformers](#)
Tyler LaBonte, Yale Song, Xin Wang, Vibhav Vineet, and Neel Joshi
WACV 2023

Selected Industry Research Experience

MICROSOFT RESEARCH Redmond, WA
Machine Learning Research Intern 2025

- Joined core development team of a 15 billion parameter multimodal vision-language model.
- Led integration effort, tool-use training, and evaluation for a multimodal computer-use agent application.
- Investigated grounding capabilities of our model on fine-grained visual reasoning tasks.

GOOGLE Sunnyvale, CA
Machine Learning Research Intern 2023

- Developed techniques to leverage Gemini LLM to improve architecture-agnostic hardware-software code design.
- Synthesized chain-of-thought and few-shot prompting strategies to generalize to data-scarce applications.

Selected Service & Awards

Reviewing: CVPR, TMLR, ICLR, ICML, NeurIPS, CVPR Workshop on Demographic Diversity in Computer Vision, ICLR Workshop on Spurious Correlations and Shortcut Learning

DoD National Defense Science and Engineering Graduate Fellowship (\$170,000) 2021

NSF Graduate Research Fellowship (\$138,000—declined) 2021